

Full Finetuning (No Adapters)

**Optimizer
State
(32 bit)**

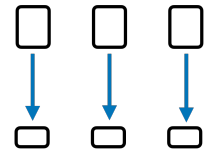
**Adapters
(16 bit)**

**Base
Model**



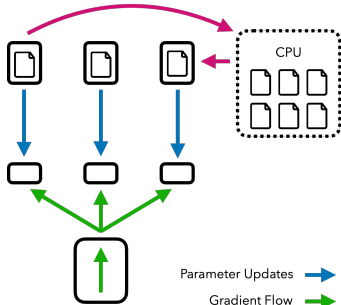
16-bit Transformer

LoRA



16-bit Transformer

QLoRA



4-bit Transformer

Parameter Updates →
Gradient Flow →
Paging Flow →